

TRAVEO™ II Family AUTOSAR FEE Release Notes

SRN223768 Version 1.11

About this document

Scope and purpose

Thank you for your interest in the TRAVEO™ II Family AUTOSAR FEE version 1.10. This document lists the installation requirements, software changes, limitations and known issues.

Intended audience

This document is intended for anyone who uses the flash EEPROM emulation (FEE) software of the TRAVEO™ II family.

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1 System Requirements and Recommendations

Software Prerequisites	Supported Version
EB tresos Studio Package for CYPRESS	26.2.0

1.1 Supported Compilers

Green Hills Software, Compiler v2017.1.4

IAR Embedded Workbench 8.0 EWARM FS 8.22.3

1.2 Supported MCAL versions (In case of install FEE with MCAL)

Only same release

1.3 Compiler Options

This section summarizes the compiler options used to build and test module. When changing the compiler options, the module must be considered untested.

Compiler	Option (Cortex®-M4F Core)
Green Hills Software, Compiler v2017.1.4	<code>-cpu=cortexm4f -thumb -thumb_lib -C99 --short_enum -align4 --no_commons --no_alternative_tokens -asm3g -preprocess_assembly_files -nostartfiles -globalcheck=normal -globalcheck_qualifiers --prototype_errors -Wformat -Wimplicit-int -Wshadow -Wtrigraphs -Wundef -reject_duplicates -c -list -Ospeed -OI -Olink -Ointerproc -Omax -fsingle</code>

Compiler	Option (Cortex-M7 Core)
Green Hills Software, Compiler v2017.1.4	<code>-cpu=cortexm7 -thumb -thumb_lib -C99 --short_enum -align4 --no_commons --no_alternative_tokens -asm3g -preprocess_assembly_files -nostartfiles -globalcheck=normal -globalcheck_qualifiers --prototype_errors -Wformat -Wimplicit-int -Wshadow -Wtrigraphs -Wundef -reject_duplicates -c -list -Ospeed -OI -Olink -Ointerproc -Omax -fhard</code>

Compiler	Option (Cortex-M4F Core)
IAR Embedded Workbench 8.0 EWARM FS 8.22.3	<code>--cpu=Cortex-M4 -e --debug -Ohs --endian=little --fpu=VFPv4_sp --no_size_constraints</code>

Compiler	Option (Cortex-M7 Core)
IAR Embedded Workbench 8.0 EWARM FS 8.22.3	<code>--cpu=Cortex-M7 -e --debug -Ohs --endian=little --fpu=VFPv5_d16--no_size_constraints</code>

System Requirements and Recommendations
1.4 Library Compiler Options

If a binary library has been delivered with this module, it has been built using the following options.

Compiler	Option (Cortex-M4F Core)
Green Hills Software, Compiler v2017.1.4	<code>-cpu=cortexm4f -thumb -thumb_lib -C99 --short_enum -align4 --no_commons --no_alternative_tokens -asm3g -preprocess_assembly_files -nostartfiles -globalcheck=normal -globalcheck_qualifiers --prototype_errors -Wformat -Wimplicit-int -Wshadow -Wtrigraphs -Wundef -reject_duplicates -c -list -Ospeed -OI -Olink -Ointerproc -Omax -fsingle</code>

Compiler	Option (Cortex-M7 Core)
Green Hills Software, Compiler v2017.1.4	<code>-cpu=cortexm7 -thumb -thumb_lib -C99 --short_enum -align4 --no_commons --no_alternative_tokens -asm3g -preprocess_assembly_files -nostartfiles -globalcheck=normal -globalcheck_qualifiers --prototype_errors -Wformat -Wimplicit-int -Wshadow -Wtrigraphs -Wundef -reject_duplicates -c -list -Ospeed -OI -Olink -Ointerproc -Omax -fhard</code>

Compiler	Option (Cortex-M4F Core)
IAR Embedded Workbench 8.0 EWARM FS 8.22.3	<code>--cpu=Cortex-M4 -e --debug -Ohs --endian=little --fpu=VFPv4_sp --no_size_constraints</code>

Compiler	Option (Cortex-M7 Core)
IAR Embedded Workbench 8.0 EWARM FS 8.22.3	<code>--cpu=Cortex-M7 -e --debug -Ohs --endian=little --fpu=VFPv5_d16 --no_size_constraints</code>

1.5 Memory Consumption(GHS)

GHS (Fee_lib.lib) Section	Size (in Bytes)
.text	33634
.bss	368
Combined:	34002

GHS (Fee_src.lib) Section	Size (in Bytes)
.text	2860
.bss	6252
.rodata	180
Combined:	9292

Note: The memory consumption of Fee_src.lib depends on the configuration.

Note: The listed memory consumption will vary depending on customer configuration.

System Requirements and Recommendations
1.6 Memory Consumption(IAR)

IAR (Fee_lib.lib) Section	Size (in Bytes)
.text	34296
.bss	345
Combined:	34641

IAR (Fee_src.lib) Section	Size (in Bytes)
.text	2900
.bss	6330
.rodata	16
Combined:	9246

Note: The memory consumption of Fee_src.lib depends on the configuration.

Note: The listed memory consumption will vary depending on customer configuration.

1.7 Explanatory Notes for this Section

Section	Description
.text	Program code
.bss	Variables that are not explicitly initialized
.rodata	Read-only data
.data	Variables with explicitly initialized values

1.8 Stack Consumption(GHS):

Function	Max Stack Usage (in Bytes)
Fee_Init	52
Fee_SetMode	56
Fee_Read	84
Fee_Write	68
Fee_Cancel	24
Fee_GetStatus	12
Fee_GetJobResult	52
Fee_InvalidateBlock	64
Fee_GetVersionInfo	48
Fee_EraseImmediateBlock	64
Fee_MainFunction	112
Fee_Clear / Fee_ClearEx	56
Fee_GetRemainingPages / Fee_GetRemainingPagesEx	52

System Requirements and Recommendations

Function	Max Stack Usage (in Bytes)
Fee_CleanupAndErase / Fee_CleanupAndEraseEx	56
Fee_SetCycleMode	64

Note: Stack consumption has been evaluated using the *gstack* utility program, which is part of the Green Hills release package. To enable the measurement of stack consumption in your project, build the source code according to the instructions given in the "Measuring Stack Consumption" section of the FEE User Guide.

Note: The listed stack consumption will vary depending on customer configuration

Note: The GHS stack consumption listed in the release notes was measured using the additional compile option "*-gs*". The GHS compiler cannot measure stack consumption for the selected optimization level (see Compilation options). Green Hills cannot exclude possible effects of "*-gs*" on optimization and stack consumption. Therefore, Infineon cannot guarantee the accuracy of these values. For more information on measuring GHS stack consumption, see the section *gstack* utility program in *Build_arm.pdf*.

1.9 Stack Consumption(IAR):

Function	Max Stack Usage (in Bytes)
Fee_Init	216
Fee_SetMode	40
Fee_Read	60
Fee_Write	52
Fee_Cancel	216
Fee_GetStatus	0
Fee_GetJobResult	24
Fee_InvalidateBlock	44
Fee_GetVersionInfo	16
Fee_EraseImmediateBlock	44
Fee_MainFunction	536
Fee_Clear / Fee_ClearEx	24
Fee_GetRemainingPages / Fee_GetRemainingPagesEx	24
Fee_CleanupAndErase / Fee_CleanupAndEraseEx	36
Fee_SetCycleMode	24

Note: To enable the measurement of stack consumption in your project, build the source code with the linker option "*--enable_stack_usage --log call_graph*". See Stack usage analysis of the IAR C/C++ Development Guide for details.

Note: The listed stack consumption will vary depending on customer configuration.

System Requirements and Recommendations

Note on “*_Bswmd.arxml”

Note that the <Module>_Bswmd.arxml files are templates that can be freely modified by the customer or RTE vendor.

These are in the *output\generated\swcd* subfolder of your *project* folder.

Named files are not tested.

1.10 Release Details

Module Software Version

1.11.x

(x=software patch version; see the delivery notes for details)

AUTOSAR Specification Version (ASR)

4.2.2

Target

MXS40

MCAL Configuration Setting

See the Resource Release Notes

Supported Derivatives

See the Resource Module Release Notes

Corresponding Fee_MemMap.h Stub File Version

1.0.0

Installation

2 Installation

See the Installation Manual for EB tresos Studio for CYPRESS AUTOSAR Software Products and Installation Manual for FEE42-TRAVEO.

3 Deviations from AUTOSAR

The following configuration parameters have properties that differ from AUTOSAR specifications.

- The value of `FeeNumberOfWriteCycles` cannot be changed. This value is automatically determined by `FeeCallCycle`.
- `FeePollingMode` is fixed value(true), so its value cannot be changed.
- `FeeNvmJobEndNotification` is fixed name, and cannot be renamed.
- `FeeNvmJobErrorNotification` is fixed name, and cannot be renamed.
- `FeeVirtualPageSize` is fixed value(4).
- The maximum value of `FeeBlockSize` is configured by `FeeBlockMaxsize`, which is a vendor-specific parameter.
- `FeeDeviceIndex` is fixed value(0).
- `FeeBlockOverhead` is fixed value(20).
- `FeePageOverhead` is fixed value(16).

Limitations

4 Limitations

- Configuration parameter: `FeeBlockNumber`

When `ConfigEx('FeeBlockConfigurationEx')` is used, the same value of `FeeBlockNumber` cannot be used between `Config('FeeBlockConfiguration')` and `ConfigEx('FeeBlockConfigurationEx')`.

Documentation

5 Documentation

User Guide for FEE is in the `\doc` subdirectory of the *installation* directory. The default location is:

`C:\CYPRESS_ESDB\Tresos26_2_0\doc`

6 Technical Support

If you have questions related to the TRAVEO™ II Family AUTOSAR FEE, contact the local support application engineer.

Version History

7 Version History

7.1 Module SW-Version 1.1

Initial module setup.

7.2 Module SW-Version 1.2

- Support TRAVEO™ II-B-E-2M.
- Support TRAVEO™ II-B-E-8M.
- Add Block check function.
- Target name is changed from CYT2 to MXS40.
- Remove MCAL modules from EB tresos help contents.

7.3 Module SW-Version 1.3

- Improved performance.
- Added automatic calculation of sector size.
- Updated Sections "Supported Compilers", "Compiler Options", "Library Compiler Options", "Memory Consumption", and "Stack Consumption" to add IAR compiler information.

7.4 Module SW-Version 1.4

- Support TRAVEO™ II-C-2D-6M support.

-
- Support EB tresos V26.2.0

[Impact]

Strict AUTOSAR specification and check for parameter configuration errors are implemented in EB tresos V26.2.0.

In addition, handling of reference paths (relative paths) such as System Description file (ARXML) is changed in EB tresos V26.2.0.

Therefore, if the current ECUC Configuration Definitions XML file contains deviations or errors, you may find errors during import to tresos26. In that case, the ECUC Configuration Definitions XML file must be modified appropriately.

In addition, if the current ARXML file contains unresolvable paths, you may find errors during import to tresos26. In that case, ARXML file must be modified.

The SW has been tested; no risks except for the low-level risk listed above were found.

7.5 Module SW-Version 1.5

- Support MISRA C:2012 coding rule

The MISRA C:2012 coding rule checks the source code.

If a deviation from the rules is required, add the deviation comment to the code and report the result.

If a deviation is for MISRA-C:2004 only, remove the deviation comment.

Version History

7.6 Module SW-Version 1.6

- Misleading comment in Module_MemMap.h

{Mip}_MemMap.h files are provided as sample template files. But, the file header comment cannot be modified, which is a contradiction. To resolve this contradiction, change the file header comment to allow user modification.

Also, to make sure that the file is not a part of the commercial product, move the *{Mip}_MemMap.h* files to the *MemMap* stub folder.

- C/NC ECC error on 32 bits read from CM7

Read access of 32 bits on a Work Flash from CM7 causes C/NC ECC error on the next 32-bit address. The work flash is always accessed 64-bit-wide when it is read by AXI. Therefore, FLS/FEE may not work properly on CM7; FLS/FEE must use DMA transfer to read the data from work flash memory. In addition, this release of the software omits ECC error output from FLS in previous versions. It now acquires the ECC error in FEE and outputs an appropriate message.

7.7 Module SW-Version 1.7-1.10

- Block data is lost in case of repetitive reset during data writing

The following problems have been solved.

If all the following conditions are met, the block ID being written will be lost.

[Pre-condition]

- 1) Reset happens twice consecutively while writing data.
- 2) First reset happens during block data (block ID is written)
- 3) Second reset happens during "start writing flag" (block ID is not written yet).

Under these pre-conditions, the block is recognized as "invalid" and will be lost on rare occasions during the next recycle timing.

- Some parameters are inconsistent between XDM and ARXM

The following problems have been solved.

The following inconsistencies are found between Fee.arxml and Fee.xdm.

- 1) SHORT-NAME is duplicated in Fee.arxml.
- 2) TS_T40D13M1I0R0 has no UUID in Fee.xdm
- 3) TS_T40D13M1I0R0 (UUID) is not described in Fee.xdm.

7.8 Module SW-Version 1.11

Resolved the following issue.

In a very rare case of all the following conditions being met, the sector data may be lost if a reset occurs when the physical sector configuration (PSC) flag is being written.

1. The FEE module creates a valid sector, and it writes the "sector valid" flag followed by the PSC flag. A power failure occurs when the PSC flag is being written.
2. At the next Fee_Init, the "sector valid" flag and the PSC flag are checked and considered as "Not Blank" by the "blank check" operation. Normal values of the PSC flag are also read out. As a result, this sector is considered as a valid sector.
3. After these two occur, data writes to the sector and power failure occur multiple times.

During Case 3, the value of the PSC flag cannot be read normally due to changes in checksum bits. As a result, the FEE module reports that no valid sector is found, which results in losing the sector data.

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